

Lab 9

In Lab Activity

Given IP Address:

192.168.1.0/27

Class C $\rightarrow$ 24 $\rightarrow$ CIDR

Step 1)

\27 $\rightarrow$ 27 bits are used in network

In IPV4, we have 32 bits

Network $\rightarrow$ 27 bits

Host $\rightarrow$ 5 bits

Step 2) Find Subnet mask

First 27 bits are 1

Last 5 bits are 0

||||| . ||||| . ||||| . |||00000

255 . 255 . 255 . 224

Step 3) How many subnets and hosts?

Borrowed bits = 3

Subnets = $2^3 = 8$ subnets

Hosts per subnet = $2^5 - 2$

= $32 - 2$

= 30

30 hosts per subnet

2 subtract for network and broadcast address

30 usable IP addresses in each subnet

Step 4) Find the block size

$$\text{Block Size} = 2^5 = 32$$

This means every subnet jumps by 32 in the 4th octet.

Step 5) List the subnets (192.168.1.X)

Subnet #	Network	Usuable	Broadcast
1	192.168.1.0	192.168.1.1-192.168.1.30	192.168.1.31
2	192.168.1.32	33-62	63
3	64	65-94	95
4	96	97-126	127
5	128	129-158	159
6	160	161-190	191
7	192	193-222	223
8	224	225-254	255